

Environmental data as a relational mediator of contentious publicness: Data curation and environmental data activism

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Abstract

This study focuses on the grassroots data practices of an environmental non-government organization (ENGO), examining the constitutive role of data in environmental activism in China. Drawing on the concept of contentious publicness, it analyses the processual and relational dynamics of the ENGO-led data activism to tackle environmental pollution at the interrelated material, spatial and temporal levels. Through participatory observations and in-depth interviews, it first examines the material agency of data infrastructure in enabling environmental participation. Then, it explores the spatiality and temporality of activists' action repertoires. The findings demonstrate that the activist engagement with data follows a non-confrontational approach, evolving in the compromised middle ground between embeddedness and marginalization. Environmental data serves as a relational mediator in activists' continuous process of making tactical responses to disrupt the status quo while not completely denying or subverting existing power relations. Moving beyond the contestational view of data, the study applies a non-binary and processual account of data activism, contributes to a deeper understanding of the relational configurations of power in data politics. It sheds light on the institutional, technological, and social imaginaries of environmentalism that shape the building of environmental data infrastructure and cultivate new forms of environmental action in the specific sociopolitical context of China. Moreover, the situated analysis of data activism contributes to diversifying the Western-centric understanding of the transformative potentials of data and calls for more scholarly attention to the relational dynamics of data politics in the Global South.

Keywords

Datafication, data activism, environmental politics, publicness, ENGOs, China

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Introduction

With the rapid development of data technologies, data serves as the new mediator of political intervention and action, enabling new forms of political activism. The tactical use of data technologies in enhancing social justice is seen as data activism (Milan and van der Velden, 2016). It is broadly understood as people's agentic engagement with data in acting upon politics and generating alternative epistemologies that can advance social change.

Environmental data has become a new force enhancing environmental justice in diverse arenas such as air, water, climate and biodiversity and across different regions of the world, reflecting its global scope and multiplicity. In

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the past decades, environmental data activism has been emerging in North America such as citizen sensing technologies applied to facilitate citizens to create counter data stories to monitor air pollution in the US (Gabrys et al., 2016), and the increasing reliance on data technologies in organising grassroots climate activism in Canada (Flawn and Soden, 2024). Similarly, in Europe, environmental politics has increasingly been shaped by data. For example, sensors are used to monitor waves of oceans in the Netherlands (Helmreich, 2009), local activists translate air pollution into actionable data in Poland (Ślosarski, 2024), and institutional and social actors collect data of birds to inform EU biodiversity policies (Ślosarski and Juchniewicz, 2024). Beyond the Western contexts, varieties of datafied environmental action occur in the Global South, including countries in Asia, South Africa, Latin America, and beyond, further exemplifying that the flows of environmental data are shaped by specific social and cultural contexts (Cinnamon, 2021; Lin, 2022; Rajão and Jarke, 2018; Sun and Yan, 2020). Local actors move away from the epistemic dominance underpinning Western imaginaries of data's transformative potential and instead engage with more situated, context-specific understandings of environmental data (Lehuedé, 2023).

Existing studies have shown environmental data activism in China often takes soft and adaptive tactics to expand the green public sphere under the state's control of civil society (Sun and Huang, 2022; Sun and Yan, 2020). However, they tend to understand data activism through bottom-up approaches that are largely shadowed by the conventional dichotomy between state and society. Though they are insightful, they have somewhat overlooked the institutional power underlying the development of environmental data activism at both material and spatial levels, which largely determines the non-confrontational nature of the state-society interaction. Moreover, the dualism between the material and the discursive also disappears in data activism. These non-binary interactions between contention and control, the technical and the social, the material and the discursive recall a new approach to unravel the entangled and relational dynamics of data politics in China.

Moving beyond the contestational views of data politics, I argue that the notion of "contentious publicness" (Kavada and Poell, 2021: 195) can be a useful framework to inform a non-binary approach that examines how relational and processual dynamics shape data activism. Kavada and Poell (2021) propose the notion of contentious publicness which considers "publicness as an interactive process" (p.191) in motion to characterize the power relations in (digital) public contention across time and space. As a concept originates in social movement studies, contentious publicness particularly addresses the relational dynamics in the process of public contention. Taking a processual perspective, it traces the emergence, circulation and diffusion of contention by looking at the interrelated material, spatial and temporal relations that have major consequences on the affordances of

(digital) media, spaces for political participation and sustainability and legitimacy of citizens' claims (Kavada and Poell, 2021).

Adopting the processual perspective, this study examines how publicness emerges through the relational dynamics of data, infrastructure, and power in the Chinese environmental sphere. This research illustrates the constitutive role of data in environmental activism by focusing on the grassroots data practices of The Institute of Public & Environmental Affairs (IPE), an ENGO in Beijing. Specifically, it explores the power relations that shape the material infrastructure and the space for the public to engage in environmental action and the sustainability of their action repertoires. Through participatory observations and in-depth interviews, this project examines the material, spatial and temporal dimensions of IPE's data activism practices to engage the public in monitoring pollution by asking the following questions:

RQ1: How do the activists engage in building data infrastructure for environmental monitoring and how do the material affordances and relations shape the process of contention?

RQ2: How do activists use data to create space for the public to participate in environmental monitoring and how does the spatial characteristics and relations shape the dynamics of data activism?

RQ3: How do the material and spatial dimensions impact the temporal characteristics such as rhythms, speed and persistence of the data activism?

Moving beyond the dualist and contentious framework, this study foregrounds a processual and relational approach to studying data activism by drawing on the notion of contentious publicness. The focus on the constitutive role of data infrastructure directs our attention to the material agency of data in environmental activism, deepening our understanding of the materiality of public contention in the age of datafication. Moreover, the contextualized study of data activism in China helps us move beyond the confines of "data universalism" (Liu and Wang, 2022; Milan and Treré, 2019; Valente and Grohmann, 2024) and foreground de-westernized understanding of the progressive potentials of data in the Global South.

The following article first reviews the literature of data activism studies, illustrating the need to move from the instrumental approach to a relational approach to studying data activism. Second, the article introduces the notion of contentious publicness, developing a processual lens to examine the relational dynamics of data activism. Third, it explains how the concept of contentious publicness can be translated into the specific context of environmental data activism in China. Then, I use IPE's data activism practices as a case to show how the processual lens informed by

the notion of contentious publicness helps us gain deeper understanding of the relational characteristics of data activism in authoritarian countries.

Data as a mediator of contention

Data activism considers “data infrastructure and data flows as a critical site of contestation” (Beraldo and Milan, 2019: 8), representing a social approach to study the politics of datafication and its sociopolitical impacts from the bottom up (Couldry and Powell, 2014). As put forward by Beraldo and Milan (2019), data are “both repertoires and stakes of contention” (p.9). Data-as-stakes is also called “data-oriented activism” which refers to contentions that take data as the goal or “objects of intervention”, while data-as-repertoires refers to “data-enabled activism” which involves contentions that are enabled through and with data (Beraldo and Milan, 2019: 6).

Applying data in advancing and innovating action repertoires to drive social change is viewed as proactive data activism (Milan and van der Velden, 2016). For instance, open data can be used to facilitate activists to meet their political goals, in which datafication serves as the new conditions that support public agency (Baack, 2015). Integrating data technologies and software into action repertoires, civil society mobilizes ordinary citizens to create their own data narratives, generate alternative knowledge and thus fostering alternative public spheres (Gutierrez and Milan, 2018). Recent studies pay attention to the materiality of data activism, exploring how counter-affordances of software and algorithms are created and how oppositional affordances enable new possibilities of action (Kazansky and Milan, 2021; Milioni and Papa, 2022). Differently, when data becomes the object of resistance, it is reactive data activism that is in play (Milan and van der Velden, 2016). MyData community’s engagement in alternative data arrangements is an example of empowering individuals in managing their own data and steering more just data arrangement (Lehtiniemi and Haapoja, 2020). Civil society groups are also active in developing tactics and strategies to defend individual rights against surveillance to promote data justice (Kazansky, 2021).

Although data activism research so far has made an insightful contribution to understanding data in social movements, it emphasizes activists’ specific data practices in political intervention, while paying limited attention to data as infrastructure and its structural power in contentious politics (Milan and Beraldo, 2024). As Beraldo and Milan (2019) note, viewing data as either stakes or repertoires risks flattening the complexity of data’s role in contentious politics, thereby obscuring the power dynamics that structure the datafied contention. Given the much more complicated data practices in various contexts, the linear understanding of data activism might be limited to capture the multiple dynamics that shape grassroots data practices. Furthermore,

Ślosarski (2023) points out that “a merely descriptive representation of assemblage composition does not fully capture the relational dynamics of power and resistance” (p.3). Therefore, a more comprehensive approach is needed to account for the profound transformation brought by datafication at both the structural and the symbolic levels.

Reflecting on how datafication changes the environment and conditions of social movements, Stefania Milan and Davide Beraldo (2024) propose the notion of “datafied movements” to move from the instrumental perspectives which mainly look at the tactical use of technology to a more comprehensive way of understanding the role of technology in political participation in datafied societies (p.271). It particularly moves our attention to “the sociotechnical mechanisms” (Milan and Beraldo, 2024: 270) underlying the production and circulation of data, advancing our understanding of the datafied environment in which the socio-technical assemblage is situated. Similarly, Ślosarski (2023) emphasizes the importance of capturing the relational struggles in the contentious politics of data in a contextual realm. He suggests taking data as relations that shape and are shaped by “its interaction with other actors and contexts” in production, transformation and usage (Ślosarski, 2023: 2). The above discussion demonstrates that there is a relational turn in data activism studies which aims to capture the multiple material-discursive dynamics through which data becomes a political actor in specific sociopolitical contexts.

Contentious publicness

The notion of contentious publicness shows strong theoretical potential in informing a processual and relational approach to analysing the multiple power relations within dynamic material-semiotic networks of data activism. Viewing contention as a process, Kavada and Poell’s (2021) conceptualization of contentious publicness highlights the relational dynamics at the interrelated material, spatial and temporal levels, taking digital infrastructure as a constituting component of public contention rather than merely an instrumental tool for social justice purposes. The notion of contentious publicness is inspiring to understanding the constituting role of data and its agency in activism because it articulates the following characteristics. First, it proposes a shift from arenas to dimensions to trace the material, spatial and temporal relations in public contention, developing a more comprehensive approach to capture the networks of power configurations and material-discursive practices (Kavada and Poell, 2021). Moving beyond the perspectives from platform studies which tend to narrow the scope to the networks and communicative process on the platform per se, this notion affirms the role of the material infrastructure in creating the environment for participation and their role in steering the dynamic process of making things public both online and offline.

It considers digital media infrastructure as a series of material arrangements around data, platforms, algorithms, and other hardware and software designing that shape the user practices and power configurations. It does not see digital media as merely a facilitator of the socio-technical assemblage but also pays attention to the power relations underneath the structural transformations brought and enabled by the material arrangements. In Kavada and Poell's (2021) conceptualization, affordances are key to the materiality and agency of digital media and thus they are important analytical factors to look at to unravel the material relations in the process of public contention. Affordances refer to a relational dynamic that sits in between users, object and the associated outcome emerging from users' interaction with the object (Evans et al., 2017). Highlighting the context in which the publicness emerges, they suggest focusing on the affordances provided by digital technology to multiple actors to trace the interaction between the material and the discursive resources. Their focus on the affordances is inspiring for the study of the material agency of data as it directs our attention to the affordances of data and the material-discursive practices that enact on both data power and human agency in the dynamic web of relations.

As Kavada and Poell (2021) suggest, the flow of publicness is not only shaped by digital infrastructure but also their spatial and temporal structure. Therefore, to fully understand the constitutive role of data in steering the process of public contention, we also need to assess the spatial and temporal configuration of publicness under the conditions of datafication. Kavada and Poell's (2021) work shows that the materiality of digital media shapes the spatial dynamics, which influences the space for the emergence of publicness. Drawing on Lefebvre's (1991) ideas about the production of social space, they focus on the interaction among the three aspects in the making of space: the perceived space (the material features of social media), conceived space (the symbolic and representational aspect of the space) and lived space (the everyday use of the space). In their conceptualization of the spatiality of publicness, they highlight the relational dynamics between designers, regulators and users of the material space. Paying attention to the relational interactions between the multiple actors, this conceptual framework provides insights into assessing the interaction between the material aspects of data, the discursive and symbolic meanings of data and the everyday use of data to open up space for participation.

Kavada and Poell (2021) also point out that the material and spatial relations consequently shape the temporality of digital media, which "affects the rhythms, speed, and degree of synchronicity of contentious communication, as well as its timing and long-term horizon" (p.197). Like space, temporality is also produced by and producing social relations and shaped by the materiality of media technologies (Kavada and Poell, 2021). Under datafied

conditions, the temporality of contentious publicness is thus closely entwined with the materiality of data and its capacity to shape people's lived experiences. Data could be considered material infrastructure that supports the architecture and its associated networks, shaping the rhythms, liveliness, durability and temporal structure of data practices. In the meantime, the temporality of data is also determined by the imaginaries and discourses inscribed in it and users' everyday practices.

Nowadays, in a datafied environment, the materiality, spatiality and temporality of publicness is integrally woven into the intertwined material and discursive processes, through which data is being produced, represented, and enacted. This shift from geographic area to dimensions enables a comprehensive approach to understanding the constitutive role of data and data infrastructure in contentious politics. This move sheds light on the interplay between materiality of data, the symbolic representation of data and human agency, and as well as the relational configuration of power at the interrelated levels. By tracing the material, spatial and temporal relations that flow in data production, data practices and meaning making, we can go deeper to explore the data-mediated configurations as the socio-technical assemblage and beyond, as the structural process. This will, in turn, deepen our understanding of the materiality of public contentions in the age of big data and how contentious publicness takes shape in datafied settings.

Second, the notion of contentious publicness emphasizes the fluid identity of participants and the flows of publicness, thus it can capture diverse forms of publicity emerging in data-mediated contentions. It challenges the limitations of dominant notions such as 'counter publics' and 'networked publics' which still confine their connotations within the western ideals of the public sphere (Habermas, 1991). Moving beyond the normative concepts of counter (publics), Kavada and Poell (2021) acknowledge the multiple forms of contestation taking place in different political systems across the globe. Thus, they turn to look at the process of contentious interaction by focusing on "(a) processes of public claim-making, (b) legitimacy of claims and claim-makers, and (c) arenas and contexts in which claims are made" (Kavada and Poell, 2021:195). The processual perspective loosens fixed identify forms and makes the notion a suitable approach to analysing the fluid and relational dynamics that shape data practices in specific contexts.

Moreover, it also opens the field of contentious publicness to authoritarian and non-democratic contexts, though it is primarily built based on empirical observations in the democratic context (Sun and Wright, 2024). With the flexibility to be employed in a non-democratic context, it provides an inclusive framework to look at activism in China, which often occur in a subtle form due to oppressive measures and does not explicitly evolve around a fixed public (Sun and Wright, 2024). Its focus on the

process of interaction rather than the episodic counter action particularly opens up the space for moderate or adaptive forms of data activism situated within China's authoritarian governance structure (Sun and Huang, 2022; Sun and Yin, 2024).

Contentious publicness in China's data environmentalism

In the Chinese environmental sphere, the nature of collective mobilizations is significantly shaped by the authoritarian context (Stalley and Yang, 2006). Green activism in China maintains nonconfrontational relations with the party-state to safeguard a safe space to promote environmental change (Wang, 2019). The non-confrontational approach has also been applied by social activists to fight against injustices in other realms such as feminists' adoption of harmless gossip to circumvent censorship when exposing sexual misconduct by male musicians in online communities (He and Huang, 2025). Through the softened tactical framing of their claims, activists carve out a middle ground between resistance and suppression. Under the authoritarian control, environmental activists and civil organizations must work within the allowed space and strategically adapt their repertoires to avoid being marginalized and oppressed by the party state. The embedded form of activism, being described as "embedded environmentalism", has proven to be effective in creating political opportunities for civil society actors to engage the public in the middle ground (Ho, 2007: 200).

Similarly, environmental data activist practices are shaped by the state's environmental policy framework and its authoritarian bureaucratic structure. The coexistence of collaboration and contestation is often found in activists' interaction with the governments, which leads to embedded data activism in strategic collaboration with the governments for legitimacy and resources support (Sun and Huang, 2022; Sun and Yan, 2020). To make their environmental data initiatives influential, ENGOs rely on local governments' disclosure of environmental information, which has been one of the core requirements of national environmental protection laws, regulations and policies since 2003 (Sun and Yan, 2020). Besides, the top-down building of data infrastructure such as pollution monitoring stations, data centres, and platforms to capture the emission data of factories has been a prerequisite for social actors to develop efficient datafied solutions to environmental issues.

In the past decades, the state has been proactively involved in designing and adopting data technologies such as data and smart technologies to trace and quantify environmental pollution and carbon emissions, as well as building data infrastructure to facilitate broader public participation in environmental politics (IPE, 2024; Sun and Yan, 2020). Meanwhile, social actors like ENGOs are

actively engaged in infrastructural projects and technological design to make raw environmental data accessible to the public and facilitate its use for advocacy (Tarantino and Zimmermann, 2017). Their political positioning and strategies to mitigate conflicts with polluting factories and governments in infrastructural building continuously shape how they leverage data in their action repertoires (Sun and Yan, 2020). As Tarantino and Zimmermann (2017) observe, environmental activists sometimes intentionally do not take the full critical potential of the data when visualizing pollution on their digital maps, adopting a compromising strategy to balance tensions with governmental actors and enterprises.

Previous studies suggest that both the design of the technological system, the material characteristics of data infrastructure and the power relations underlying the socio-technical systems impact the data flows and the process of contention. However, less scholarly attention has been paid to the material relations formed in the building and maintenance of the data infrastructure and how the material arrangements and material-discursive networks shape the dynamics of data activism so far. Furthermore, data environmentalism in China involves complex interaction between the state and ENGOs and is shaped by the relational dynamics of various actors include data infrastructure, designers, users, activists, NGOs, enterprises and governments (Sun and Huang, 2022; Sun and Yan, 2020). In the authoritarian context, the dichotomies between state and society, resistance and control, grassroots data practices and institutional infrastructural building disappear. Moreover, the less contentious distinctions between the technical and the social, the material and the discursive also fade. While the current scholarship on contentious politics of data is very helpful to understand the varieties of data activism, it may not be sufficient to fully capture the complexity and fluidity of the relational dynamics of environmental data activism in China.

Moving beyond the dualist views of data activism, this study draws on the notion of contentious publicness to trace the interrelated material, spatial and temporal relations in data-mediated contention to capture the relational struggles for environmental justices in the authoritarian context of China. Nevertheless, the original focus of contentious publicness on public claim-making and its legitimacy needs to be reoriented to account for the process of data-mediated contention since it unfolds in different ways in China. Specifically, the process of making data public, make claims through data and gaining legitimacy for the environmental data activism is embedded in the power relations produced through networked data practices that are not conventionally confrontational (Sun and Huang, 2022; Sun and Yan, 2020). Sometimes, obtaining access to data and gaining legitimacy for the tactical use of data relies on collaborative relations with the government. So, it is a dynamic process that comes into being through relations, networked

Table 1. The interviewee list.

Interviewee number	Role in IPE
1	Founder of IPE and Blumap App
2	IT Director of IPE
3	Senior Research Officer of IPE
4	Software Designer, Data Analyst
5	Green Chain Officer
6	Data Product Manager
7	Communication Officer

practices and power configurations, rather than in an explicit form of public claim or contestation. Thus, this study suggests the need to extend the connotations of contentious publicness by underscoring the material-semiotic networks that highlight the interaction of the materiality of data, human agency and the associated material-discursive practices within the configured power networks to better understand the emergence of publicness in the authoritarian context.

Methodology

The case: IPE's digital maps

The Institute of Public & Environmental Affairs (IPE) is an ENGO established in 2006 in China. Its primary mission is to encourage government agencies, corporations, and the public to take an active role in environmental protection. IPE has designed the Air Pollution Map as part of its efforts to track the enterprises' pollution data and contribute to the global fight against climate change. These digital tools provide detailed, real-time data on pollution, allowing users to monitor corporate pollution behaviours and hold companies accountable for their actions. Being a high-profile data activism initiative for environmental action, IPE's data practices serve as a timely case study for understanding the role of data in grassroots environmental activism within authoritarian regimes.

Data collection and analysis

First, this involves participant observation of IPE's daily work by visiting their offices and engaging in their public activities (the annual press conference). The participant observation provides deeper insights into their methods of gathering pollution data, the technical processes involved in transforming the data into actionable information, and their efforts to facilitate public use of the environmental data. Additionally, it also offers a valuable opportunity to learn about how IPE interacts with governments, enterprises, and media to create and maintain a space for public participation in their data campaigns.

Second, semi-structured interviews with IPE members including IT staff, public relations officers, other involved staff and relevant government officials are conducted. The interviews cover the following topics: a) the role of the interviewees in IPE; b) data initiatives IPE has organized and their role in those activities; c) their ambitions for and opinions of those data initiatives; d) how they interact with authorities, enterprises, media and the public to open up and sustain a participatory space. As the state's control over civil society and social actors intensifies, access to official voices becomes increasingly constrained for researchers. During my attempt to interview a government official, I only received a highly scripted response, which was not included in the data because it was unlikely to provide meaningful insights. With other interview data (Table 1), I analyse the visions and imaginaries behind IPE's data initiatives, exploring the nuanced power negotiations among multiple actors and how that shape the data-mediated environmental action.

Third, relevant information about IPE's data activities is collected from their website, app, social media accounts (Weibo and WeChat), and media reports. The textual data supplements the participant observation and interviews, enabling a deeper understanding of IPE's efforts to collaborate with media and leverage social media to mobilize the public in monitoring pollution.

Combining the multiple sources of data, a reflexive thematic analysis is done to assess how participatory affordances of environmental data are imagined, how material-discursive practices around data unfold in everyday spaces, and how networked power relations shape these processes. Based on the thematic analysis, the study explores the material, spatial and temporal aspects of IPE's data activism practices.

Materiality of environmental data: Imagining participatory affordances

As discussed above, affordances are important factors to explore to understand the materiality of publicness under datafied conditions. IPE plays the role of "critical data designer" (Fortun et al., 2016: 3), imagining the affordances of the data infrastructure and shaping how raw environmental data can become accessible, searchable, usable and actionable for the public in the sociopolitical context of China. As data activists imagine and enable the agentic possibilities of data, software and platform, the affordances of environmental data are materialized. Moreover, the production and construction of environmental data is not only a material process but also involves political and social work, which consequentially shapes the affordances and use of the data.

Holding the belief that open data is a potential solution to tackle environmental pollution, IPE has been building the database to publicize environmental pollution in water

and air since 2006. The database was originally named “pollution map” in its early days. Since environmental data is considered as politically sensitive in China, IPE mainly gathers data from governments and EPAs to mitigate the contentiousness of the database. To engage the public in environmental participation, *accessibility* of governmental data is a key precondition. However, due to the sensitivity of the pollution data, “governments only disclose a very limited amount of environmental data in 2000s; thus, another key part of IPE’s job is to advocate for enhanced governmental openness and data disclosure” (Interviewee 1). Thus, gaining governmental support and maintaining collaborative relationships with governments is always important political work for IPE.

However, “the central government and local governments hold different attitudes to their work”, as interviewees mentioned. The state and the Ministry of Ecology and Environment (MEE) have been supportive of information disclosure (Interviewee 3), since environmental protection has been given priority in Xi Jinping’s agenda. However, the local environmental protection agencies (EPAs) are reluctant to make the local pollution data open as they are afraid of being blamed by the higher-level governments for the problem of smog and environmental pollution (Interviewee 1). Due to the uneven policy implementation between the central and local levels (Spire, 2011), the central and local governments face varying degrees of pressure from grassroots activism (Hsu, 2017). Therefore, IPE adopts different measures to maintain relationships with governments at various levels.

Since the central government has the regulatory power to manage NGOs in China, the recognition from the state is important for IPE to obtain legitimacy and political capital, which secures their position in advocating for changes in environmental laws and regulations to promote information disclosure. Thus, IPE has been making efforts to foster trusting relationships with the central government through both formal and informal channels to gain state support. Informal networks, such as co-hosting conferences and engaging with authorities at academic conferences, have been maintained to bridge the gap between the government and NGOs. For instance, IPE frequently participates in conferences jointly organized by universities and governments to network and build trust with authorities. In April of 2019, Mr Jun Ma, IPE’s funder, was invited to speak at the conference on big data-driven governance and environmental protection organized by Tsinghua University (China Daily, 2019). His speech on the use of big data in environmental monitoring was echoed by Mr Shijin Liu, the Deputy Director of the Economic Affairs Committee of the 13th National Committee of the Chinese People’s Political Consultative Conference (CPPCC) and the Deputy Director of the Development Research Centre of the State Council and Mr Jinglian Wu, Research Fellow at the Development Research Centre of the State Council.

And they all emphasized and elaborated on the need to advance the application of data-driven technologies in environmental governance.

Embracing the linkage with NGOs, state institutions open up collaborative opportunities for IPE to engage in environmental governance. In addition to maintaining the informal networks with IPE, the governments have been actively incentivizing IPE to take a greater role in addressing environmental problems. Since 2019, Jun Ma has been designated as a Special Observer by the Minister of Ecology and Environment, and he was awarded as one of the most influential people in ecology by the central government in 2020 (Ecology China Net, 2020). Using the data disclosed by governments, IPE designed the Bluemap app to visualize environmental pollution, which was awarded the “Best App” alongside other governmental apps by Esri China (Beijing) Co. Ltd in 2018 (Sohu News, 2018). The symbolic recognition from the central government represents the state’s support of IPE’s work, with which IPE is able to expand the space of collaboration into local level. For the state, the symbolic interaction with IPE or ENGOS enhances its legitimacy and the public trust towards the party-state (Song, 2024a).

This collaborative relationship with the state is reciprocal, with the state’s reliance on ENGOS as “service providers” or “helpers” in its environmental governance bureaucracy (Dai and Spire, 2018: 64). Therefore, the central government holds a favourable attitude towards IPE’s work, working to strengthen the partnerships with IPE and achieving closer collaboration. The state gives IPE the permission to develop the Bluemap app by using environmental data gathered from governments is a sign of state support, which lays foundation for strategic collaboration between IPE and governments. For the state, it is an opportunity to outsource the task of pollution monitoring to IPE, whose database facilitates governments in monitoring and regulating polluting factories (Interviewee 2 and Interviewee 3). For IPE, the symbolic support from the central government legitimizes its data work, which can facilitate strategic collaboration with local governments on a larger scale. As a matter of fact, IPE has been strategically driving local governments to open the access of environmental data to the public. In 2008, IPE created the Pollution Information Transparency Index (PITI) to assess the performance of local governments in information disclosure, leveraging the discursive power of transparency to encourage greater data openness. With the PITI, IPE evaluates, scores and ranks the local EPAs of more than 120 major cities annually (IPE, 2024) and the results of the assessments are made accessible to both the public and higher-level governments through press conferences and media coverage (Interviewee 3). The ranking serves as an effective tool to encourage local governments to improve information disclosure under the pressure of stringent central inspections between 2016 and 2020 (Interviewee 1 and Interviewee 3).

With the goal of making the data accessible to the public, IPE has created the Bluemap website and app as a national platform to standardize the procedures for data collection, aggregation, analysis, and visualization, and enhancing the searchability, usability, and visibility of the data. Moreover, the national platform, built with the central support of the state, also facilitates data sharing by local governments (Interviewee 4) to ensure the accessibility of environmental data (Interviewee 3). To make data agentic for environmental change, IPE's work pertains to both the inner operation of the app but also the sociopolitical context for its use to carve out the context-specific affordances of the data. For instance, they have designed software to unify the diverse formats of regional data collected from local EPAs across the country. Applying the standardized data format, IPE's IT department further cleans and crafts the data in a way that they can be easily inserted into the mobile app and then make the interfaces user-friendly, facilitating citizens' everyday use of the data to monitor pollution (Interviewee 4).

The standardization of data formats and the design of the search function of the app simplifies the procedures citizens need to go through to check the pollution data. As demonstrated by Figure 1, the app affords the *searchability* of different types of environmental data, including specific monitor stations (image in the left), trends in pollution in a specific geographical area (the image in the middle), and wind data (the image in the right). Turning emission data into various types of searchable information enables users to develop their own knowledge about the sources of pollution, the human and natural factors affecting air quality and the geographical impact of pollution. IPE's technical work plays an important role in keeping the smooth flow of data, shaping them into app data, and ultimately making pollution searchable, mappable, and publicly visible via the Bluemap app.

Recognising the representational capacities of data, IPE has been making efforts to increase the *visibility* of the data in various ways to problematize environmental pollution in the wider public realm. For example, they have creatively turned data into narratives depicting environmental pollution in cities, which are made publicly available and searchable on the platform. Figure 2 shows various narratives of the air quality in Hotan (Xinjiang Province), informing citizens the average PM2.5 in the recent 12 months, its ranking among 338 cities across China, and the yearly record of polluting days which are visualized by colour-coded indicators. The visual representations of pollution data from various perspectives help construct discourses about the heavy pollution in Hotan in 2025, enhancing the public understanding of the local environmental situation.

Moreover, the Bluemap app streams data disclosed from governmental monitoring stations across municipal, provincial, and national levels, forming a GPS-based

national air map, where citizens can find the real-time pollution data and practice cross-territory monitoring of the polluting factories (Figure 3). All these material arrangements to datafy and visualize environmental pollution enhanced the *usability* of the data, laying the foundation for the public to further participate in environmental action.

To mitigate the contentiousness of the environmental data, IPE has engaged in significant social and political work through collaborative networks with both the central and local governments. Their imagined affordances of the data infrastructure are contextualized in China's particular socio-political relations and the state's rationale of governance. IPE's creation and curation of the environmental data reflects their contextual understandings of participatory affordances, which promotes the accessibility, searchability, usability and visibility of the environmental data. However, it needs to align with the legal framework and the state's ideologies of environmental governance. Consequently, the contextual imagination of affordances and material features legitimizes and enables the agentic use of the data infrastructure.

Spatial and temporal relations: The flows of data-mediated publicness

IPE activists develop their action repertoires to mobilize public participation in pollution monitoring as a form of material-discursive intervention, exemplifying how data is imagined and used by the wider citizenry across space and time. Curating the environmental data in a way that facilitates civic action is an important part of IPE's work. IPE's sociopolitical choices also play an important role in ensuring that their material-discursive practices open up a stable space for participation and have tangible effects in the long run. As Law (2008) suggests, it is the networked configuration of power relations that produces the durability and stability of the material-semiotic practices. In China, socio-political and organizational arrangements greatly affect the space, the scale, the rhythm and the persistence of NGOs' data practices.

To ensure both sustainability and a safe space, IPE must undertake socio-political work to integrate their data activities into existing bureaucratic organizations. The networks and mutual trust built between IPE and governments are essential for extending the spatial and temporal scale of their data activities. The central government needs IPE to help gather local data that local governments are reluctant to release due to economic concerns (Interviewee 3). This political need from the state opens up opportunities for IPE to crowdsource grassroots data from citizens to help the central government stay informed about the facts of environmental pollution across the country. With the citizen data, the central government can target the polluting areas and

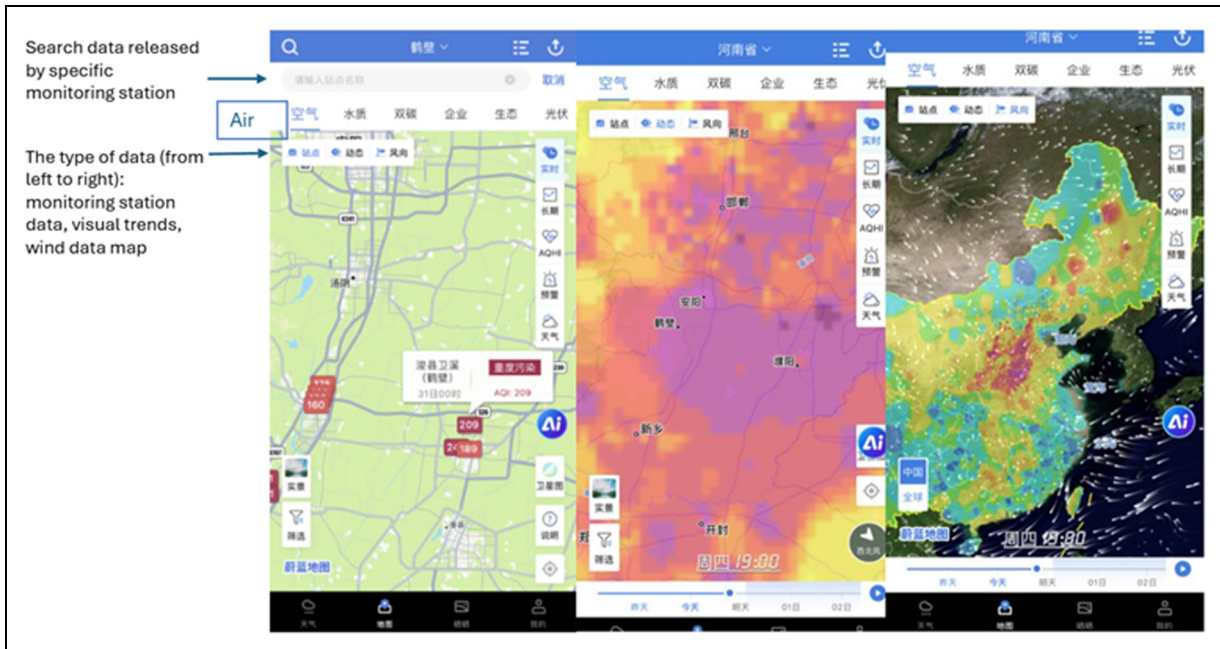


Figure 1. The interface and the search function of Blumap app.

pressure the local governments and factories to reduce pollution.

To overcome the data access obstacles caused by the inefficient transparency of the local environmental data, IPE has designed a reporting function on the water map of Blumap to allow users to report black and odorous water bodies on the app in August 2016. The establishment of this crowdsourcing and reporting system via Blumap app is a result of the strategic collaboration between IPE, Ministry of Housing and Urban-Rural Development, and Ministry of Ecology and Environment. In 2015, the central government released the document “Guidelines for the Remediation of Urban Black and Odorous Water Bodies”, providing instructions for local governments and social organizations to address the pollution of urban water bodies (IPE, 2018). The guidelines emphasize that citizen data serves as an essential reference point in the identification and investigation of black and odorous water bodies as well as the evaluation of the effectiveness of measures taken to address water pollution. Thus, in February 2016, the two ministries jointly launched the ‘Urban Water Environment Public Participation’ reporting channel on WeChat, a widely used social media platform, encouraging the public to report information about foul and filthy waters. Seeking broader public input as a collective effort to assist the central government in addressing water pollution, the two ministries outsourced the promotion of the public participation in monitoring water pollution to IPE and ‘Clear Water as Neighbour’ ENGO networks. Consequently, the public

monitoring data from Blumap app is linked to the official reporting platform on WeChat.

IPE has been training its app users, in collaboration with other ENGOs (Friends of Nature), to report black and smelly rivers in a clear and standardized format, ensuring that all relevant data is captured in a meaningful and actionable way. Figure 4 shows the suggested reporting format for citizens to effectively document pollution incidents to facilitate the authorities to take appropriate action and demonstrates that citizen’s reporting is responded by the relevant governmental agents with detailed explanation about the work they are doing to sort out and clean up the river. The reported data from citizens plays an important role in identifying the polluted rivers that were not on the official list (IPE, 2016). Although the unofficial data effectively contributed new data sources and thus enhanced the transparency of the water data, the bottom-up monitoring by the public and ENGOs has been suppressed by some local authorities, who intervene to demand that they cease monitoring and reporting (IPE, 2018).

In addition, IPE has launched another data initiative, *micro-reporting* aimed at mobilizing the public to monitor and report factory pollution through its app and social media exposure in the wider public realm. The operation of micro-reporting heavily relies on the IPE’s structured curation of enterprises’ pollution data on Blumap app and its infrastructural integration with other social media platforms. As Figure 5 shows, citizens can navigate the digital map of enterprises to access the environmental

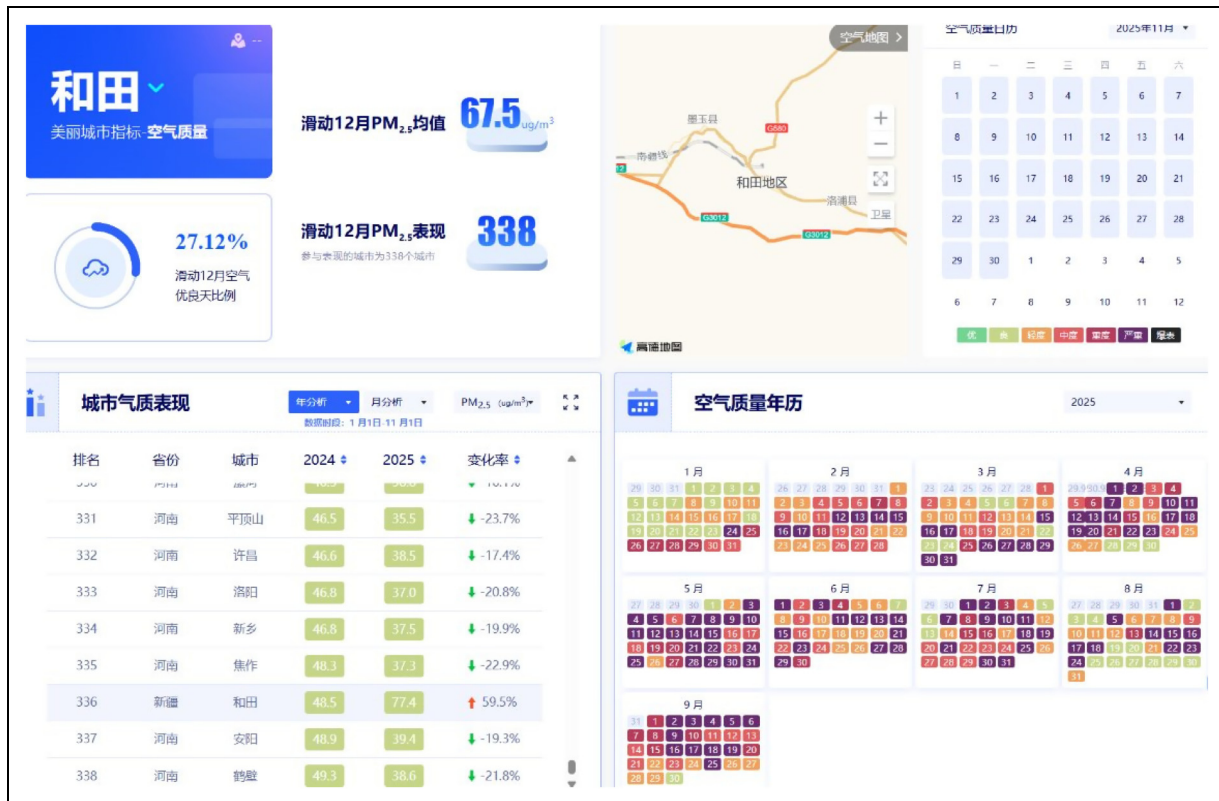


Figure 2. Visualizations of AQI data in Hotan, Xinjiang province.
Source: <https://www.ipe.org.cn/CityEnvironment/Air.aspx?cityid=401>.

data of factories near their GPS locations, which clearly marks the polluting factories in red. When citizens click the factory, they will be led to the page that presents the specific environmental data with information about the specific pollutants or substances that are exceeding the allowed emission limits. If citizens want to pressure the factory to reduce pollution, they can press the share button in the top right corner to post the pollution data on Weibo, WeChat Moments, WeChat group chats, and Tencent QQ Zone, exposing their polluting behaviour to a wider network. Moreover, activists can tag the local EPAs' official Weibo accounts to notify them about the reported pollution and request responses (Figure 6). The operation of data-mediated reporting relies on activists' tactical use of the affordances of different platforms (Blumap app, Weibo, Wechat, QQ space), which created a wider space for grassroots participation.

However, the tangible effects of data practices associated with *micro-reporting* rely much on institutional arrangements. Local EPAs play a decisive role in holding factories accountable for their pollution, especially large enterprises that consistently exceed emission standards (IPE, 2022). Realizing the significance of institutional support from local governments, IPE has pursued strategic collaboration with the local state in organizing data-mediated *micro-reporting*,

aiming to establish feedback loops between the public and the government. Moreover, local governments and EPAs rely on support from ENGOs and the public to stay informed about factory pollution, using this information as evidence for legal enforcement and supervision while mitigating tensions between enterprises and the governments (Interviewee 2). For instance, local governments in Shandong province proactively seek help from IPE in monitoring enterprises and sharing public complaints with them under pressure from the central government. To ensure that the public reporting data can effectively prompt factories and local EPAs to address public concerns, a three-level response system has been established in Shandong. This system operates through the vertical interaction of official Weibo accounts at the provincial, municipal, and county levels, forming the institutional foundation for the public complaints and supervision. However, the three-level response system merely proves the success of the strategic collaboration between Shandong governments and IPE. For most local governments, evidence-based micro-reporting poses a significant threat, as they fear it could lead to penalties from the central government, especially in the context of central inspections on local environmental governance. Therefore, it is essential for IPE to maintain strong networks with concerned citizens and other

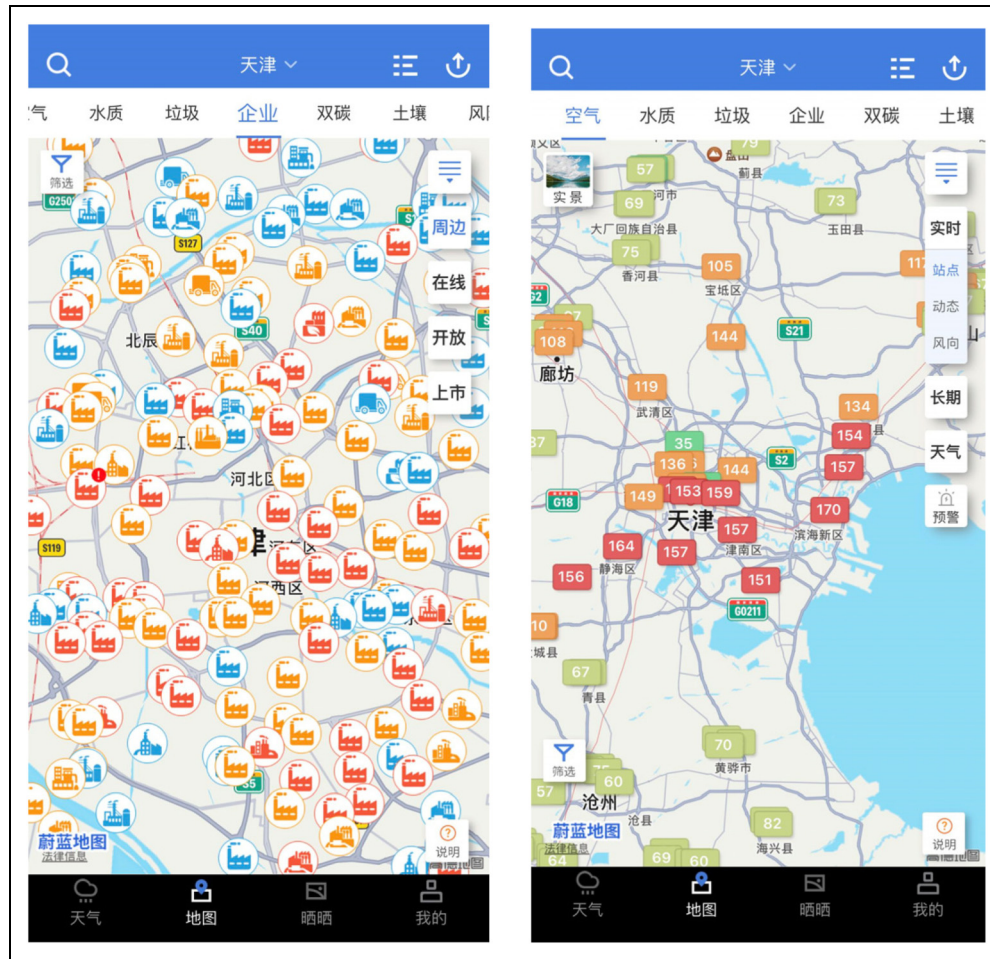


Figure 3. Air map showing factory pollution (screenshot of content from the app).

NGOs to form an *Alliance of Green Environmental Organizations*, practicing data-based oversight and counterbalancing the possible oppression from local governments (Interviewee 1). With organizational and institutional support, this model of material-discursive intervention turns from an ad-hoc basis to a sustained model, expanding the space of environmental participation.

The stability of the *micro-reporting* system not only relies on IPE's curation of pollution data and its spatial organization across various platforms but also the temporal relations. The disclosure of real-time pollution data is the key determinant of the temporal characters of IPE's data activism practices. All app users including citizens, NGOs, enterprises and authorities receive automated notification of the real-time data and are immediately informed of any updates. This enables them to translate the data into situated knowledge of the environmental issues and offer public accounts of pollution on a daily basis. Using the real-time data to support their claims channelled through the *micro-reporting* system, citizens can enhance the legitimacy of the public scrutiny of real-time pollution, thus pressuring

the governments to respond to the public complaints efficiently. The opening of real-time data allows the NGOs, citizens and the governments to see the problem, making environmental pollution visible in both the public realm and the institutional sphere. The data initiatives show that IPE's curation of data on the platform enables public participation in environmental monitoring across space and time. The material-discursive intervention gains its spatial reach and temporal endurance through the configurational stability generated by multiple and intersecting interactions among heterogeneous actors including the data infrastructure, IPE, central and local governments, and other NGOs. Its durability lies less in fixed structures than in the relational process that continually (re-)configures and sustains the material-discursive assemblages.

Discussion

Tracing IPE's activist engagement in building the data infrastructure and curating data on the platform to enable public monitoring, environmental data serves as a relational

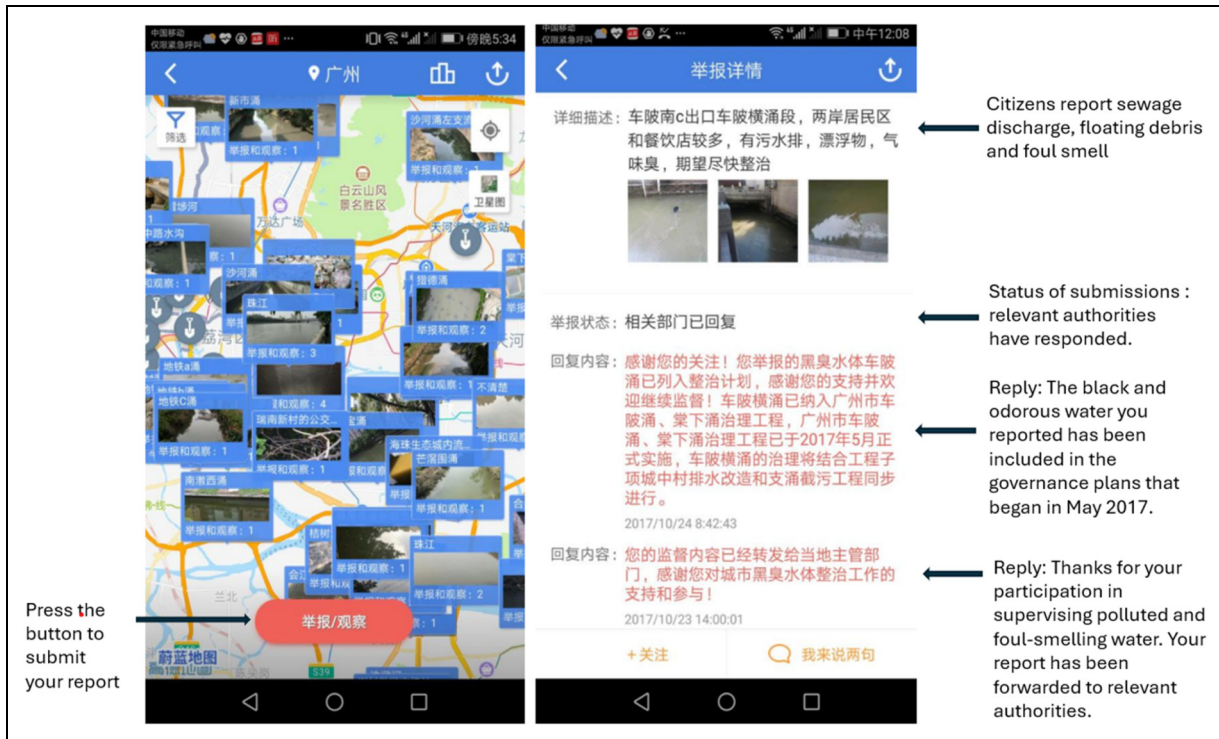


Figure 4. Reporting pollution related to a river in the city of Guangzhou. (Sourced from IPE report on Observing Hundreds of Foul & Filthy Waters, 2018).

mediator in the situated material, spatial and temporal process. It is not only the materiality, spatiality and temporality of environmental data but also the networked power relations underpinning the sociotechnical assemblage and data flows that shape the process of contention. Moving beyond the conventional dichotomy of control and contention, IPE's data activism practices unfold through relational dynamics that privilege non-confrontational interaction between state and society. They employ strategic ambiguity and tactical moderation to sustain their work within what Wang (2019: 39) describes as the "massive middle ground" between embeddedness and marginalization. IPE has been positioning itself as a service provider for the central government and establishing partnerships with the governments through symbolic and strategic collaboration, practicing outsourced environmental data activism. The partnership with the central government provides IPE with an institutional veneer that creates legitimacy and outward political opportunities to expand the influence of their data work at the material, spatial and temporal levels.

At the material level, IPE strives to make the Bluemap app accessible, searchable, usable and actionable for the public to engage in environmental action through their contextualized imagination of the material affordances of environmental data. Establishing trusting relations with authorities and facilitating collaboration among multiple actors, IPE intervened in environmental politics by providing

contextual affordances for different stakeholders to address their own needs. In this case, the affordances of the app are not in an oppositional or confrontational nature, as revealed in most of the data activism projects in western countries (Kazansky and Milan, 2021; Milioni and Papa, 2022). Instead, IPE's design choices embody contextual affordances, maintaining the critical potential of data to some extent while also moderating their tactics to sustain trusting relations with both central and local governments. So, the platform mediates a continuous process of making tactical responses to disrupt the status quo while not completely denying or subverting the existing political system. Rather than crafting counter affordances consistent with conventional contentious view of data politics, the agentic affordances of data are contingently imagined in the specific socio-political context of China.

IPE's data infrastructure is not only material but also performative, thus consequently shaping the spatiality and temporality of their activist data practices by mediating how pollution data is visualized and enacted in the digital space over time. As suggested by Law (2008), the scale of material-discursive intervention grows, when the material-semiotic networks extend. Through strategically networking and collaborating with both central and local governments, IPE has created new socio-technical spaces for the public to engage in evidence-based monitoring by linking its app to wider social media ecosystems such as Weibo

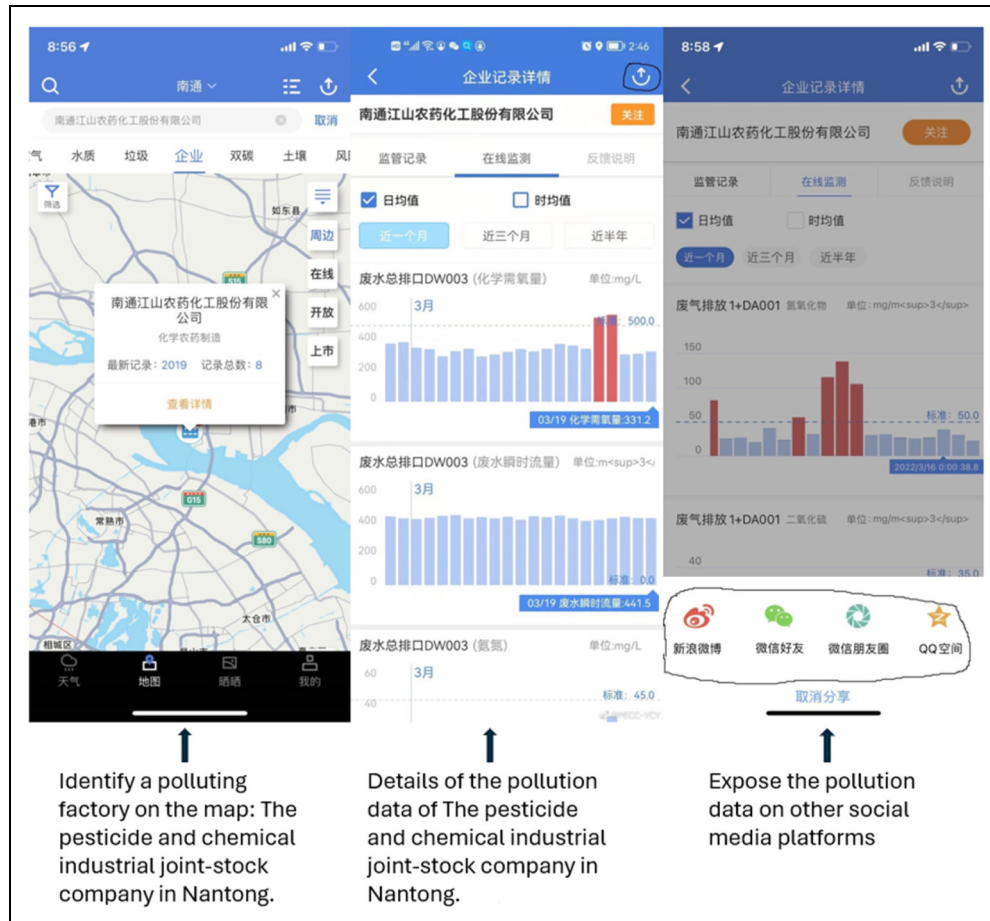


Figure 5. Steps for reporting factory pollution using the app data (screenshot of content from the app).

and WeChat. The cross-platform linkage between Bluemap and popular social media platforms extends the circulation of environmental data and facilitates environmental participation in a broader networked space.

Temporally, the articulation of environmental data within these material-semiotic networks shapes the rhythms of data activism through the cycles of monitoring and reporting, and prompts responses from authorities. The automatic real-time alerts stimulate a quick monitoring and response system, enhancing the efficiency and sustainability of environmental governance. Furthermore, the algorithmic aggregation of real-time data into monthly and yearly representations of regional pollution produces longitudinal knowledge about environmental problems in particular cities or provinces, enabling local governments to formulate more targeted environmental policies and interventions. Within the multiple spatial and temporal structures, contentious publicness emerges through dynamic material-semiotic networks. It is the various configurations of environmental data that enact pollution through varying material-discursive practices across space and time, producing multiple spatial and temporal modes of visibility, action, and response.

As discussed above, the materiality, spatiality and temporality of the data-mediated publicness is shaped by the networked power relations beneath the flows of data. Obviously, the data configurations and the material-semiotic assemblages are influenced by the political negotiations between IPE and the governments in China, which reflects flexible paternalism in the state-NGO interaction. The study finds the mutual accommodation between the state and ENGOS, laying the basis for their collaborations. First, the state has shown flexibility in regulating and governing NGOs, with the increased institutional space for service-oriented NGOs (Song, 2024b). Through both symbolic and strategic collaboration with IPE, the central state can easily access public and grassroots data about pollution as evidence to pressure local governments to address environmental pollution more effectively. Positioning themselves as a service-provider for the governments, IPE wins state support in developing their data initiatives, which grants them political legitimacy to extend the space of their participatory projects at the local level. Furthermore, the fragmentation inherent in central-local administrative system and contingent factors (Qiaoan, 2020) open a window

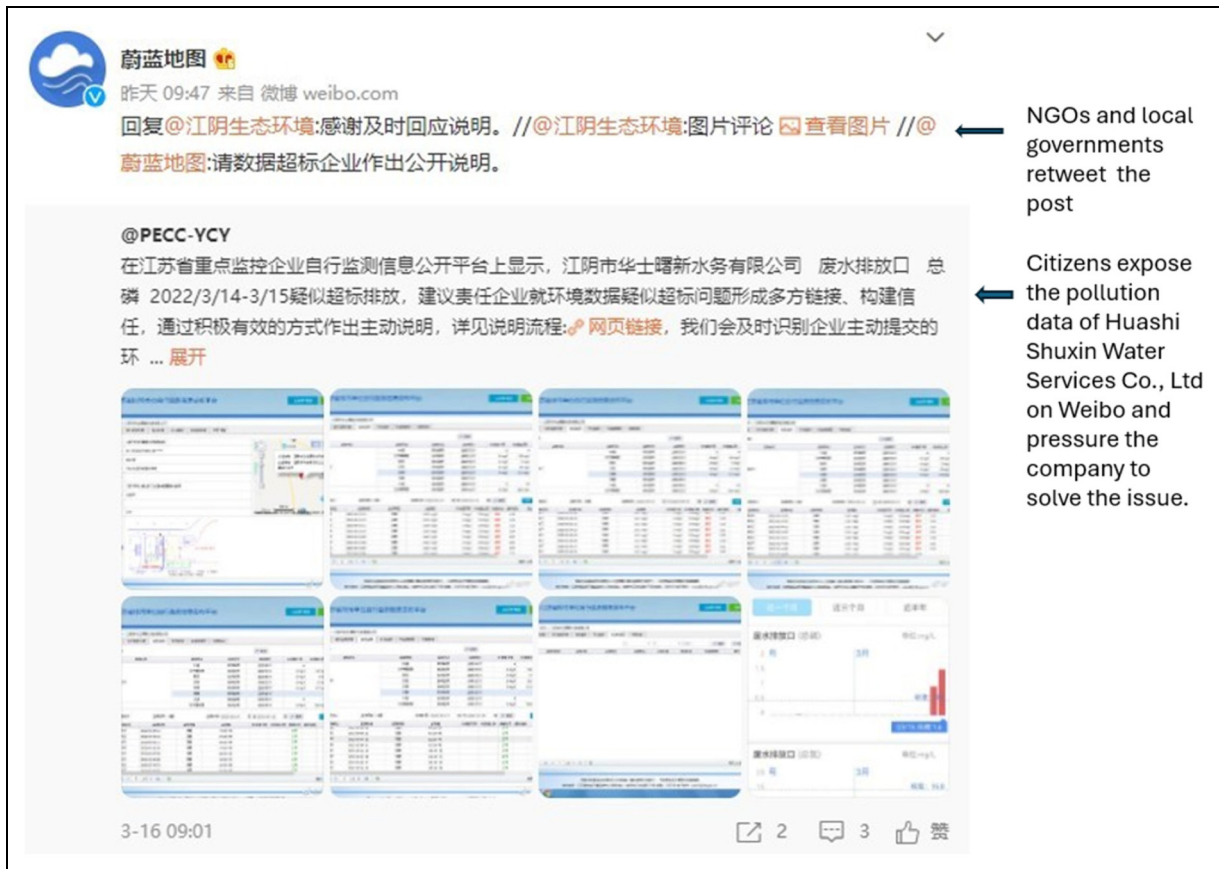


Figure 6. The interaction between citizens, ENGOs and local EPAs (screenshot of Weibo posts).

of opportunities for IPE to push their agenda forward in the political sphere and thus penetrating the space of the state.

However, the mutual accommodation between the governments and IPE is asymmetrical. IPE's data work reflects the de-politicalization of the space for ENGO-led data activism in China. As Lee and Zhang (2013) suggest, the authoritarian domination of the state is a "non-zero-sum situation", and it allows room for bargaining and negotiation (p.1475). Yet, the state dominates the scope of the space for negotiation. In this case, IPE has proactively sought to build and maintain collaborative relations with the governments due to its dependence on the state for legitimacy. Thus, the state can take it as an opportunity to extend itself into the space of ENGOs which enables the state to maintain its control over the civil society in a subtle way. While the state moves from direct control to inclusion, the boundaries of the institutional space opened for IPE are still largely shaped by the governments.

Conclusion

Drawing upon the concept of contentious publicness, this study takes a processual approach to understand data activism as a relational process that unfolds at the material,

spatial and temporal levels. This research contributes to the current scholarship in the following three aspects. First, as the study suggests, the boundaries between contention and control, state and society are not fixed at the inter-related material, spatial and temporal scales. Instead, they are contingent and stabilize through constant interactions within the relational material-semiotic networks. The non-linear dynamics of data activism, especially in the authoritarian context of China, challenges the dichotomy between control and resistance, suggesting the need to shift from the contentious politics of data to the relational configuration of data, which echoes Ślosarski's (2023) proposal to take data activism as relational dynamics.

Second, paying attention to the data assemblages and the dynamics of contentious publicness, the study sheds light on how data becomes constitutive of the formation of contentious publicness in civic practices around and through data. It contributes to expanding the framework of contentious publicness which is originally formulated in alignment with the characteristics of social media infrastructure and social media use. By highlighting the networked and multiple configurations of data, the study underscores a need to pay more attention to contextualized affordances of data and the relational dynamics of material-semiotic

networks to better understand the formation of contentious publicness in the age of datafication.

Third, the analysis reveals that civil society actors use non-confrontational or soft tactics to interact with state actors to sustain a compromised middle ground for bottom-up data practices and mobilizations in China. This finding reaffirms that data activists tend to stick to “strategic universalism” and apply “tactical localism” (Barreneche and Lombana-Bermudez, 2023: 3652) in the Global South, recognising both the shared normative aspirations and the need to situate their data practices in the specific socio-political context. It contributes to diversifying the predominantly Western-centric debates on the contentious politics of data and calls for continuous efforts to incorporate indigenous perspectives to better understand the relational dynamics of data politics in the Global South.

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